Dylan McDermott

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WORK EXPERIENCE

Weather & Catastrophe Analyst January 2025-Present

Property & Liability Resource Bureau, Remote

- Build and maintain production Python and ArcPy systems that ingest and validate NOAA, IEM, radar, observation, warning, and model data; generate daily maps, archives, exports, email content, and ArcGIS updates.
- Process hundreds to thousands of reports, files, and features per day while reconciling late reports and detecting duplicates, missing timestamps, invalid identifiers, incomplete location fields, and silent download failures.
- Develop claims-facing ArcGIS Experience Builder applications for hail, wind, hurricanes, and current weather, including custom REST API integrations for date filtering, layer control, map synchronization, and location-specific reports.
- Evaluate new datasets for scientific defensibility, spatial and temporal resolution, bias, provenance, update behavior, and appropriate interpretation; distinguish observations from modeled and remotely sensed estimates.
- Automated recurring production tasks that previously required hours of manual work while keeping maps, tables, exports, and downstream records synchronized.

Field Sensing Research Associate (Contract) June 2024-December 2024

Corteva Agriscience, Johnston, IA

- Co-developed and operated the Smartstick, a wheeled field-sensing platform that collected thousands of timestamped, Trimble GPS-linked canopy and air-temperature measurements per walk across seven research sites.
- Independently built the Python and ArcPy pipeline from raw tables through visual quality control, outlier and GPS filtering, point creation, inward plot buffers, spatial joins, maps, tables, shapefiles, and plot-level summaries.
- Automated approximately one hour of processing per collection - roughly 100 hours across the campaign - and applied one reproducible workflow across sites.
- Compared field measurements with drone imagery, LiDAR-derived canopy structure, treatments, and crop-stress response to validate spatial patterns and research findings.
- Co-designed and built a 16-chamber automated soil-gas system, a multi-instrument reference weather station, and tracking dashboards for more than 200 deployed infrared radiometers.

Graduate Research Assistant May 2022-May 2024

Iowa State University, Ames, IA

- Designed and ran WRF and Noah-MP experiments at multiple resolutions to quantify how U.S. land-use change affected Midwest rainfall, surface fluxes, moisture transport, and mesoscale convective systems.
- Prepared model inputs by translating CESM/LUMIP land-use data to WRF and Noah-MP grids and evaluating present-day versus 1850 vegetation scenarios.
- Processed and compared ERA5, CESM, WRF, GOES, MODIS, MRMS, radar, and observational datasets using Python, xarray, MATLAB, NumPy, and NetCDF.
- Produced geospatial analyses, derived fields, maps, scientific visualizations, technical writing, and presentations for interdisciplinary research.

Undergraduate Research Assistant May 2021-August 2021

Iowa State University, Ames, IA

- Conducted boundary-layer meteorology research by identifying cases, analyzing observational and gridded atmospheric datasets, and producing geospatial visualizations to interpret physical mechanisms.

SKILLS

- Geospatial data engineering and spatial analysis
- Python and ArcPy workflow automation
- Raster/vector processing and map production
- Data quality control, validation, and reconciliation
- Remote sensing, drone imagery, and LiDAR
- REST API integration and operational monitoring
- Meteorological and catastrophe analysis
- Radar, GOES, MODIS, MRMS, and NWP data
- WRF/Noah-MP modeling and simulation
- Field sensing and GPS-linked instrumentation
- Machine learning and AI-supported analysis
- Technical communication and uncertainty analysis

SOFTWARE

- Python, SQL, MATLAB, R, C++, FORTRAN
- ArcGIS Pro, ArcPy, Enterprise/Server, Experience Builder
- NumPy, xarray, OpenCV, Jupyter, NetCDF, AWS
- Linux, Windows, Git, GR2Analyst

EDUCATION

Iowa State University, Ames, IA January 2022 - May 2024

Master of Science, *Meteorology* GPA: 3.52/4.0

- Thesis: Impacts of U.S. Deforestation on Rainfall from Mesoscale Convective Systems

Iowa State University, Ames, IA August 2018 - May 2022

Bachelor of Science, *Meteorology*

- Senior Thesis: Extreme Convective Wind Magnitude and Duration in Iowa
- Dean's List honors

PERSONAL PROJECTS

AI Weather Intelligence Platform 2025-Present

- Developing a full-stack platform that integrates 200+ weather and environmental data streams into source-backed historical, current, and forecast analyses.
- Builds automated evidence workflows and user-facing geospatial products that translate complex environmental data into maps, plots, reports, and actionable answers.
- AWS 10,000 AIdeas Competition Semi-Finalist.

AWARDS

Esri Special Achievement in GIS (SAG) Award 2025

- Contributed to PLRB catastrophe analytics and ArcGIS applications recognized for innovative use of GIS technology.